

AMRUTHA M

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EDUCATION

- B.Tech(Hons) in Computer Science, RV University, Bengaluru, Karnataka
 - PU-2, Deeksha Center for Learning, Bengaluru, Karnataka
 - 10th Grade, RNS International School, Bengaluru, Karnataka
- CGPA: 9.63, 2022 - 2026**
Percentage: 97.5, 2022
Percentage: 93.8, 2020

WORK EXPERIENCE

- Backend Intern, Hyperface** *Bangalore, Jan 2026 – May 2026*
- Implemented **RSA-based encryption** for authentication flows, developed and made changes to key backend services including customer lookup, transaction APIs, and core banking integrations.
 - Migrated services to **Kafka-based asynchronous processing** with dedicated topics for webhooks, card events, and bulk operations; authored integration/unit tests and resolved production issues.
- Intern, Broadrange AI** *USA (Remote), August 2024 – February 2025*
- Built an **AI-powered RFP Response Generator** using **OCR (Tesseract)**, **SpaCy**, and **Llama 3.1**, automating proposal drafting and structured information extraction.
 - Developed a full-stack workflow for reviewing generated responses and implemented **PDF generation** using ReportLab for export-ready proposals.
- Project Intern, NOKIA** *Bangalore (Remote), June 2024 – September 2024*
- Developed an automated vulnerability analysis pipeline using **Grype** and **Syft** to generate SBOMs and identify vulnerable Go dependencies with remediation details.
 - Built a dependency upgrade framework leveraging web scraping and the **Libraries.io API** to recommend and apply secure package versions automatically.
- Project, IBM** *Bangalore (Remote), January 2025*
- Analyzed enterprise compliance datasets to generate actionable insights and automate reporting workflows, reducing manual effort.
 - Developed a web-based dashboard for compliance monitoring, report visualization, and stakeholder access to key metrics.

PROJECTS

- AutoAssess:AI Powered Quiz Generation and Self-Assessment System** — *Spring Boot, Java, MySQL, Kafka, REST APIs*
- Developed **AutoAssess**, a full-stack AI-powered assessment platform using **React, Spring Boot, Spring Security, JWT, JPA, MySQL, Apache Kafka, and REST APIs**, enabling users to upload documents, automatically generate quizzes using LLM integration, attempt assessments, receive instant scoring, and review detailed performance history.
 - Designed an **event driven architecture with Apache Kafka**, implementing asynchronous communication between modules through document upload and score publishing events. Built analytics and notification consumers, integrated JWT-based authentication and authorization, DTO-based API design, assessment analytics (average, best, worst scores), notification management, and secure user-specific data access using Spring Security Context.
- Solar Flare Detection using Deep Learning** — *Deep Learning, Python, TensorFlow*
- Built a solar flare detection system on labeled solar image datasets using a deep CNN (13M parameters) with Conv2D, BatchNorm, MaxPooling, Dropout, and Dense layers, along with preprocessing, normalization, and augmentation.
 - Achieved 96.2% accuracy, 84.2% precision, and 80.1% recall, validated through a confusion matrix and inference on unseen solar imagery.
- MemoryGraph AI: A Dynamic, Context-Aware Hybrid Knowledge Graph Framework for Documents** — *FAISS, SentenceTransformer, RAG*
- Designed and implemented a **Hybrid GraphRAG** architecture using **Neo4j AuraDB, SentenceTransformers (all-MiniLM-L6-v2)**, where LLM extracted entities and relationships are persisted as a **multi-document knowledge graph** and enriched with **384-dimensional vector embeddings** for semantic retrieval.
 - Developed a **context-aware query engine** that performs **Neo4j vector index search, Cypher-based graph traversal, cross-document entity linking**, and **graph based LLM answer generation**, enabling multi-hop reasoning, persistent knowledge memory, explainable graph evidence, and hybrid retrieval.

RELEVANT COURSEWORK AND SKILLS

- **Programming Languages:** C++, Java, Fundamentals of Python
- **Core Concepts and Skills:** Object Oriented Programming, Operating Systems, Data Structures, Database Management System, JavaScript (Basics),MySQL, Springboot

ACHIEVEMENTS AND ACTIVITIES

- Secured 3rd place at a **national-level hackathon** organized by IEEE RV University for building a smart academic collaboration platform powered by ML-based recommendations and efficient search using tries.
- Awarded **scholarship** by the university for **academic excellence(CGPA),ranking in top 5%** of students
- **Club Founder and President(2023-24)– ChipTech:** Organized , lead techincal events for the club
- **Open source contributions in Hacktoberfest 2024 and GirlScript Summer of Code 2024**
- Solved 700+ questions on leetcode which helped in enhancing my problem solving skills.